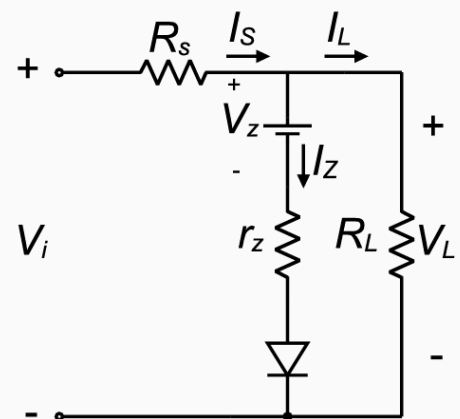
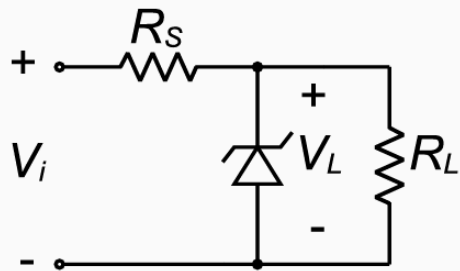


Voltage Regulator

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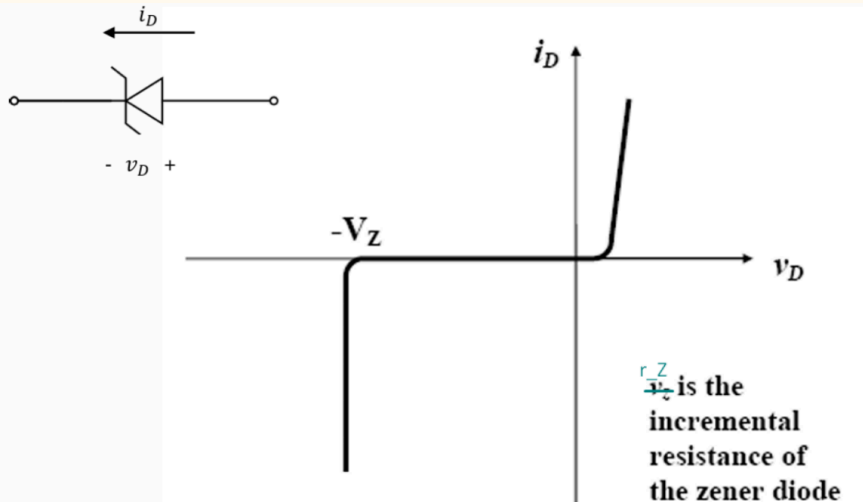
Zener Shunt



- ※ R_s required to limit current
- ※ min current required = max load current + breakdown current
- ※ Need some way to drop voltage

$$V_L = V_i \frac{r_z}{R_s + r_z} + V_Z \frac{R_s}{R_s + r_z} - I_L (R_s // r_z)$$

For $R_s \gg r_z$, $V_L \approx V_Z$



※ Above is from Thevenin the Zener shunt without load, then add in the load

※ Alternative is to do KCL and write V_L in terms of V_{in}

※ The model is both for AC and DC, DC voltage = V_Z (battery) + $r_z i_D$

$$\Delta V_L = S_V \Delta V_i + R_o \Delta I_L + S_T \Delta T$$

$$S_V = \left. \frac{\Delta V_L}{\Delta V_i} \right|_{\substack{\Delta I_L=0 \\ \Delta T=0}} \quad \text{Input regulation factor,}$$

$$R_o = \left. \frac{\Delta V_L}{\Delta I_L} \right|_{\substack{\Delta V_i=0 \\ \Delta T=0}} \quad \text{Output resistance,}$$

$$S_T = \left. \frac{\Delta V_L}{\Delta T} \right|_{\substack{\Delta I_L=0 \\ \Delta V_i=0}} \quad \text{Temperature coefficient.}$$

Smaller the better

$$S_V = \left. \frac{\Delta V_L}{\Delta V_i} \right|_{\substack{\Delta I_L=0 \\ \Delta T=0}} = \frac{r_z}{R_s + r_z}$$

$$R_o = \left. \frac{\Delta V_L}{\Delta I_L} \right|_{\substack{\Delta V_i=0 \\ \Delta T=0}} = -(R_s // r_z)$$

$$\frac{\Delta V_L}{\Delta V_Z} = \frac{R_s}{R_s + r_z}, \quad \Delta V_i = 0, \quad \Delta I_L = 0$$

$$S_T = \frac{\Delta V_L}{\Delta T} = \frac{\Delta V_L}{\Delta V_Z} \frac{\Delta V_Z}{\Delta T} = \frac{R_s}{R_s + r_z} \theta_{ZT}$$

Change in temperature affects the Zener diode junction voltage.

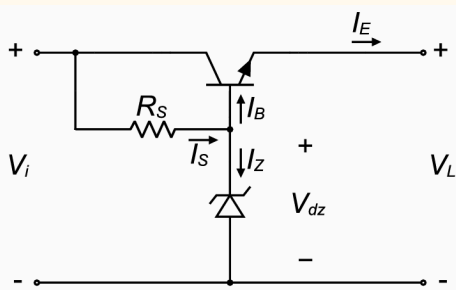
$$VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%$$

- Voltage (load) regulation
- Change in load voltage under no load and full load
- full load is minimum expected load R

$$LR = \frac{\Delta V_L}{\Delta V_i} \times 100\%$$

- change in load voltage against line (input) voltage
- line regulation, normally measured in no load

Series Pass Regulators

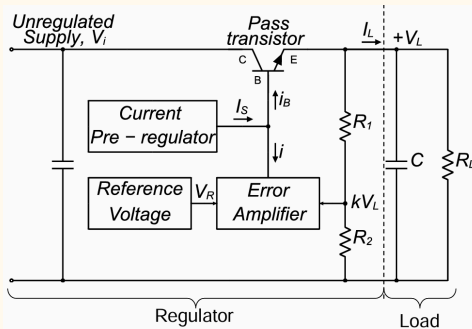


$$I_E = I_{ES}(e^{V_{BE}/V_T} - 1)$$

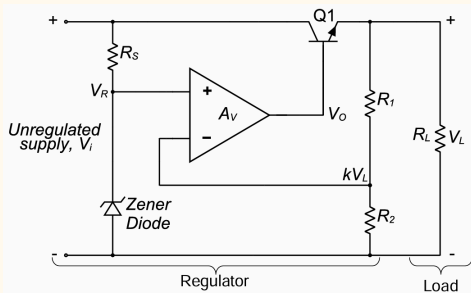
$$V_T = \frac{k_B T}{e}$$

- ✖ Its a CC amplifier with the input and bias set by the zener
- ✖ Output will be around zener - 0.7V drop (~ unity gain)
- ✖ The ideal for this configuration is to avoid drawing too much current from the zener, making the r_z affects the circuit less
 - the transistor isolates the current, the output load current is from the power supply, not the zener
 - factor of β better
- ✖ R_s is needed to biasing both the transistor base current and zener current, to design:
 - starts from max load current
 - divide by beta, to get the maximum required base current
 - then add in the zener breakdown current
 - R_s should be chosen to have enough current flow to supply both
 - supplying more current is ok, it changes the reference voltage a little bit, just don't burn too much power
- ✖ Changes in the input voltage is mainly dropped by the power transistor, but the current both zener will increase, hence the line regulation is not perfect
- ✖ V_{BE} is strongly dependent on the temperature also

With Negative Feedback



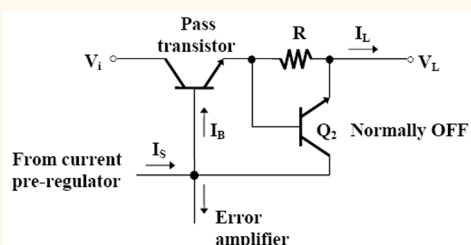
- ✖ Load current will no longer affect the reference voltage unlike the zener series pass transistor case
- ✖ the error amplifier acts like a buffer for the reference voltage
- ✖ Current pre-regulator is to bias current for the transistor and the error amplifier, so the error amplifier is not sole providing the current to the base
- ✖ To ensure the transistor is not saturation (must be in forward active mode) the input to output drop out cannot be smaller than $V_{CE, SAT}$, should give some buffer also since the input is not stable $\Rightarrow V_{in} - V_{out} \geq 10 * V_{CE, SAT} \approx 2V$
- ✖ Also called linear regulator, power dissipation = $I_L (V_{CE})$



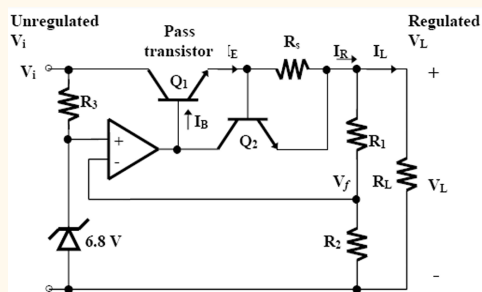
- ✖ Negative feedback forces the sampled voltage to be same as zener reference

$$V_R = k V_L \quad k = \frac{R_2}{R_2 + R_1} \quad V_L = V_R / k$$

- ✖ Choosing R1 and R1
 - big enough to draw negligible current compared to laid
 - small enough compared to op amp input impedance



- ✖ Due to negative feedback, if output is shorted, it will turn the transistor fully on, passing large current, leading to kablooeey
- ✖ Current sense resistor is used to bias the Q2 V_{BE} , design based on max current limit of Q1 $\Rightarrow$ choose $R = 0.6/I_{max}$
- ✖ The protection happens when $I = I_{max} \Rightarrow V_{BE}$ of Q2 is enough
 - can think of Q2 diverting the bias current away from the Q1 base or shunts the V_{BE} of Q1
 - when protection triggers it will just hold it at around the I_{max}



※ Choose current sense resistance to be $V_{BE(cut-in)}/I_{max}$

※ Choose zener biasing resistor to pass enough current at minimum input voltage but not blow up at maximum input voltage

※ Minimum β Q1 relies on the (max collector current)/(max op amp output current)

※ Make sure power rating of everything is ok

※ Putting the feedback after the R_s , negative feedback will account for the voltage drop across the R_s . The load regulation is still good at the cost of needed a bigger dropout